I (1 mark)

1. ${}^nP_n = n!$

Explanation: ${}^nP_r = \frac{n!}{(n-r)!}$ here $r = n$

$\therefore {}^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!}$

$0! = 1$

2. If $|A| = m$ and $|B| = n$ then $n(A \times B) =$ ——

Answer: $n(A \times B) = mn$

3. If $f : A \to B$ is a function from A to B then A is called ——

Answer: Domain

4. If $(x^2 - 1, y + 3) = (3, 4)$ find values of x and y.

Answer: $x^2 - 1 = 3$

$x^2 = 3 + 1$

$x^2 = 4$

$x = \pm 2$

$y + 3 = 4$

$y = 4 - 3$

$y = 1$

Explanation: If two ordered pairs are equal then their corresponding elements should also be equal.

5. Define injective function

Answer: A function $f: A \to B$ is said to be one-one (injective) if different elements in A have distinct images in B.

6. State Pigeon hole Psinciple.

Pigeon hole Psinciple states that if n items are put into m containers, and $n > m$, then at least one container must contain more than one item.

7. If a student can choose a project either 20 from mathematics or 35 from CS or 15 from Engineering. In how many ways a student can choose a project.

Answer: $20 + 35 + 15 = 70$ ways a student can choose a project

(The Rule of Sum used here)

II (Three Marks)

1. Find the number of permutations of the letters of the word INDEPENDENCE.

Answer:

. Total number of letters = 12 letters

out of which N appear - 3 times
E appear - 4 times
D appear - 2 times

Therefore,

number of arrangements $= \frac{12!}{4!\,3!\,2!}$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \quad 3 \times 2 \times 1 \quad 2 \times 1}$$

$$= 1\,663,200$$

2. Let $f : R \to R$ where $f(x) = 3x + 7$ for all $x \in R$.
Prove that f is one-one.

Answer:

A function f is one-one if 2 different inputs have different outputs i.e If $f(a) = f(b)$ then $a = b$.

So to prove one-one we can assume that two outputs are equal and then show that the corresponding inputs must be equal.

$f(x) = 3x+7$

Assume $f(x_1) = f(x_2)$

$3x_1 + 7 = 3x_2 + 7$

$3x_1 = 3x_2$

$x_1 = x_2$

Since $f(x_1) = f(x_2)$ implies $x_1 = x_2$
function satisfies the condition for a one-one function

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3. If $A = \{1, 2, 3, 4\}$ give an example of a relation on A which is reflexive and transitive, but not symmetric.

Answer: Given, $A = \{1, 2, 3, 4\}$

→ For a relation to be reflexive, it must contain all the self-pairs: $(1,1), (2,2), (3,3), (4,4)$

→ For a relation to be symmetric if $(a,b) \in R$ then $(b,a) \in R$
Now to make it not symmetric we can add an ordered pair in one direction: eg: add $(1,2)$ but do not add $(2,1)$

Now relation $R = \{(1,1), (2,2), (3,3), (4,4) (1,2)\}$
↘ Reflexive, not symmetric

→ A relation is transitive if $(a,b) \in R$ $(b,c) \in R$ then $(a,c) \in R$.
Here $(1,2) \in R$ and $(2,2) \in R$ also $(1,2) \in R$
and $(1,1) \in R$ and $(1,2) \in R$ also $(1,2) \in R$

Therefore the Relation $R = \{(1,1) (2,2) (3,3) (4,4) (1,2)\}$
is reflexive, transitive but not symmetric.

4. Let set $X = \{1,2,3,4\}$ and relation $R = \{(x,y) \mid x > y\}$ Draw the graph of R and also give its matrix.

Answer:

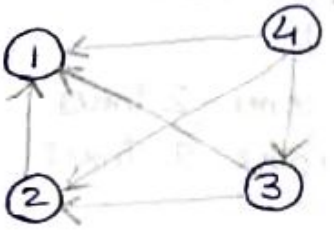
Given $X = \{1,2,3,4\}$

$X \times X$ (cross product) $= \{(1,1)(1,2)(1,3)(1,4),(2,1),(2,2),(2,3),(2,4)$ $(3,1),(3,2),(3,3),(3,4),(4,1),(4,2),(4,3),(4,4)\}$

relation $R = \{(x,y) \mid x > y\}$

So $R = \{(2,1),(3,1),(4,1),(3,2),(4,2),(4,3)\}$

Graph Representation of R



Matrix Representation of R

	1	2	3	4
1	0	0	0	0
2	1	0	0	0
3	1	1	0	0
4	1	1	1	0

5. In how many ways can the letters of the word 'MATHEMATICS' be arranged such that the vowels must always come together. ?

Answer:

MATHEMATICS

vowels in the word } = A E A I → 4 letters out of which A comes 2 times

vowel arrangements $= \frac{4!}{2!}$

Treating vowels as one letter

M T H M T C S (A E A I) → total no: of letters = 8 letters

here M come 2 times
T come 2 times

Therefore no: of different ways to arrange:

$$= \frac{8!}{2!\,2!} \times \frac{4!}{2!} = \frac{8\times7\times6\times5\times4\times3\times2\times1}{2\times1\;\;2\times1} \times \frac{4\times3\times2\times1}{2\times1}$$

Total letters, vowels

$$= 120960$$

6. An urn contains 15 balls, 8 of which are red and 7 are black. In how many ways can 5 balls be chosen so that i) all the five are red
ii) 2 are red and 3 are black
iii) Atleast 3 are red

b

Explanation:

In this question we will use combinations because we are choosing balls, and the order in which they are chosen does not matter.

The combination formulae is: $^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$

where n - total no: of objects
r - no: of objects you want to select

i) all the five are red

There are 8 red balls, and we need to choose 5 red balls, so $n = 8, r = 5$

$$\therefore {}^{8}C_{5} = \frac{8!}{5!\,(8-5)!} = \frac{8!}{5!\,3!}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1 \quad 3 \times 2 \times 1}$$

$$= \underline{56}$$

ii) 2 are red and 3 black

Here we have two separate choices

→ choose 2 red balls out of 8 red $\quad n = 8, \; r = 2$

$${}^{8}C_{2} = \frac{8!}{2!\,(8-2)!} = \frac{8!}{2!\,6!} = \frac{8 \times 7 \times 6!}{2 \times 1 \times 6!} = \frac{56}{2}$$

$$= \underline{28}$$

→ choose 3 black out of 7 black $\quad n = 7 \quad r = 3$

$${}^{7}C_{3} = \frac{7!}{3!\,(7-3)!} = \frac{7!}{3!\,4!} = \frac{7 \times 6 \times 5 \times 4!}{3 \times 2 \times 1 \times 4!}$$

$$= \underline{35}$$

Choosing 2 red ball and 3 black ball:

$$= {}^{8}C_{2} \times {}^{7}C_{3}$$

$$= 28 \times 35$$

$$= \underline{980}$$

iii) Atleast 3 are red.

"atleast 3" means 3 or more

since we are choosing exactly 5 balls, the possible set of combinations are
3 red, 2 black
4 red, 1 black
5 red, 0 black

we calculate each case separately and add them.

case 1: 3 red and 2 black

choose 3 from 8 red and choose 2 from 7 black

$$^8C_3 \times {}^7C_2$$

$$\frac{8!}{3!(8-3)!} \times \frac{7!}{2!(7-2)!}$$

$$\frac{8!}{3!\,5!} \times \frac{7!}{2!\,5!} = \frac{8\times7\times6\times5!}{3\times2\times1\times5!} \times \frac{7\times6\times5!}{2\times1\times5!}$$

$$= 56 \times 21 = \underline{1176}$$

case 2: 4 red and 1 black

choose 4 from 8 red and choose 1 from 7 black

$$^8C_4 \times {}^7C_1 = \frac{8!}{4!(8-4)!} \times \frac{7!}{1!(7-1)!}$$

$$= \frac{8!}{4!\,4!} \times \frac{7!}{1!\,6!}$$

$$= \frac{8\times7\times6\times5\times4!}{4\times3\times2\times4!} \times \frac{7\times6!}{6!}$$

$$= 70 \times 7 = \underline{490}$$

case 3: 5 red and 0 black

choose 5 red from 8 red & choose 0 black

$$^8C_5 \times {}^7C_0 \qquad \because {}^7C_0 = 1$$

$$\frac{8!}{5!(8-5)!} = \frac{8!}{5!\,3!}$$

$$= \frac{8\times7\times6\times5!}{5!\,3\times2\times1} = \underline{56}$$

Now add all cases since all are different possible cases

$$1176 + 490 + 56 = \underline{\underline{1722}}$$